

Assignment : Pose Based Activity Recognition Project Report

Name: Areeb Abdul Ghani

Roll No: 23-AI-76

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Department: BS Artificial Intelligence

University: Dawood University of Engineering & Technology

Abstract

Computer Vision is an important field of Artificial Intelligence that enables computers to understand and analyze images and videos. In this project, a human activity recognition system was developed using pose estimation and video processing techniques. The system uses MediaPipe Pose to detect body landmarks and calculate joint angles for identifying activities such as running and sitting.

The project also includes video merging, frame processing, and annotation techniques using OpenCV. The detected activities are displayed in real time on the processed video. This system demonstrates how computer vision can be used in surveillance, fitness tracking, healthcare monitoring, and sports analytics.

1. Introduction

Computer Vision is a branch of Artificial Intelligence that enables computers to extract meaningful information from digital images and videos. Human activity recognition is one of the major applications of computer vision, where a system detects and classifies body movements.

In this project, a pose estimation model is used to recognize body posture and movement. Pose estimation helps in identifying different body joints such as shoulders, elbows, knees, and hips. By calculating angles between joints, activities can be classified automatically.

This project specifically focuses on:

- Human pose detection
- Joint angle calculation
- Activity classification
- Video annotation and processing

The implementation is performed using Python libraries such as OpenCV, MediaPipe, NumPy, and Matplotlib.

2. Objectives

The main objectives of this project are:

1. To detect human body landmarks using MediaPipe Pose.
 2. To calculate body joint angles for posture analysis.
 3. To classify human activities such as running and sitting.
 4. To process and merge video frames.
 5. To annotate detected activities on video output.
 6. To demonstrate the application of computer vision in motion analysis.
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3. Tools and Technologies Used

The following tools and libraries were used in this project:

3.1 Python

Python is used as the main programming language because of its simplicity and extensive support for machine learning and computer vision.

3.2 OpenCV

OpenCV is used for:

- Reading video frames
- Video processing
- Frame resizing
- Drawing annotations
- Saving output videos

3.3 MediaPipe

MediaPipe Pose is used for:

- Detecting human body landmarks
- Tracking body joints
- Pose estimation

3.4 NumPy

NumPy is used for:

- Mathematical operations
- Array manipulation
- Angle calculation

3.5 Matplotlib

Matplotlib is used for visualization and plotting.

4. Methodology

The project follows the following workflow:

Step 1: Video Input

The system first loads the input video. Frames are extracted from the video for processing.

The video contains human activities such as running and sitting.

Step 2: Pose Detection

MediaPipe Pose model is applied to detect body landmarks.

The detected landmarks include:

- Shoulder
- Elbow
- Wrist
- Hip
- Knee
- Ankle

These landmarks are important for understanding human posture.

Step 3: Joint Angle Calculation

Joint angles are calculated to identify movement.

For example:

- Knee angle
- Elbow angle
- Hip angle

These angles help determine whether the person is sitting, standing, or running.

The angle is calculated mathematically using landmark coordinates.

Step 4: Activity Classification

A rule-based classification approach is used.

The classification is based on body posture and movement:

Running Detection

If the body posture and movement angles change rapidly, the activity is classified as **Running**.

Sitting Detection

If knee and hip angles indicate a bent posture, the activity is classified as **Sitting**.

Step 5: Video Annotation

The detected activity label is displayed on the video frame.

For example:

Activity: Running

or

Activity: Sitting

This makes the output video more understandable and user-friendly.

5. Implementation Details

The project is implemented using Python programming.

The major tasks include:

Importing Libraries

Different libraries such as OpenCV, MediaPipe, NumPy, and Matplotlib are imported.

Model Initialization

MediaPipe Pose model is initialized for body landmark detection.

Frame Processing

Each frame of the video is processed individually.

Pose Landmark Detection

Body keypoints are extracted from each frame.

Angle Computation

Mathematical formulas are used to compute angles.

Activity Recognition

Conditions are applied to classify activity.

Output Generation

The processed video is saved with annotations.

6. Results and Discussion

The project successfully detects human body posture and recognizes activities.

Observed Results:

- Human body landmarks were detected accurately.
- Running activity was identified successfully.
- Sitting posture was recognized.
- Video annotations displayed real-time labels.

Challenges Faced:

Some issues were observed during execution:

1. Incorrect pose detection in fast motion.
2. Landmark detection failure in unclear frames.
3. Running person not detected properly in some cases.

Despite these limitations, the system produced effective results.

7. Applications of the Project

This project has many real-world applications:

Healthcare

Patient movement monitoring.

Sports Analytics

Tracking athlete posture and movement.

Surveillance

Monitoring suspicious activities.

Fitness Tracking

Exercise posture analysis.

Smart Systems

Human behavior understanding.

8. Future Improvements

The project can be improved by:

- Using deep learning models
 - Increasing detection accuracy
 - Adding more activities such as walking, jumping, and dancing
 - Real-time webcam integration
 - Multi-person detection
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9. Conclusion

This project demonstrates the use of computer vision for human activity recognition using pose estimation. MediaPipe Pose successfully detects human body landmarks, while OpenCV processes video frames efficiently.

The system identifies activities such as running and sitting using joint angle calculations and rule-based classification. Although some pose detection limitations exist, the project successfully achieves its objectives and highlights the importance of computer vision in real-world applications.